H-B-00581A (GSA-FSS)
January 15, 1970
INTERIM REVISION OF
Fed. Spec. H-B-581
August 15, 1947

INTERIM FEDERAL SPECIFICATION

BRUSHES, SHOE

This Interim Federal Specification was developed by the Standardization Division, Federal Supply Service, General Services Administration, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

The General Services Administration has authorized the use of this Interim Federal Specification as a valid exception to Federal Specification H-B-581, dated August 15, 1947.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers two types of brushes used primarily in the care of shoes, one for applying polish and the other for polishing.

1.2 Classification.

1.2.1 Types. Brushes covered by this specification shall be of the following types, as specified (see 6.1):

1.2.2 Classes. Brushes covered by this specification shall be of the following classes, as specified (see 6.1):

Class 1 - Horsetail hair.
Class 2 - Mixture of horsetail hair and synthetic filament.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issues in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

H-S-951 - Synthetic Filaments for Brushes, General Specifications for.
R-N-91 - Naphthalene, Technical.
QQ-W-461 - Wire, Steel, Carbon (Round, Bare, and Coated).
PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-636 - Box, Fiberboard.
PPP-B-665 - Boxes, Paperboard, Metal-Stayed (Including Stay Materials).
PPP-B-676 - Boxes, Setup.

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).
Fed. Test Method Std. No. 191 - Textile Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, amd Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

FSC 7920

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(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices In Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Washington.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specification procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards:

ASTM D 276-T - Methods of Identification of Fibers for Testing Textiles.
ASTM D 2016 - Test for Moisture Content of Wood.
ASTM D 2130 - Method of Identification of Wool and Other Fibers by Microscopic Examination.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

Commercial Standard:

CS 35-61 - Hardwood Plywood

(Copies of Commercial Standards, which are Department of Commerce publications, may be obtained upon application accompanied by money order, coupon, or cash to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

National Classification Board:

National Motor Freight Classification.

(Application for copies shall be addressed to the American Trucking Association, Inc., Attention: Tariff Order Section, 1616 P Street N. W., Washington, D. C. 20036.)

Uniform Classification Committee:

Uniform Freight Classification.

(Application for copies shall be addressed to the Uniform Classification Committee, 202 Union Station, Chicago, Illinois 60606.)

(Technical Society and Technical Association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Materials.

3.1.1 Horsetail hair. The hair in class 1 brushes shall be 100 percent horsetail hair, unflagged, undyed, unbleached, and sterilized by one of the procedures described in 4.2.1.1. The average diameter of the horsetail hair shall be not less than 0.0055 inch and single measurements shall be not less than 0.0045 and not more than 0.0092 inch when determined as specified in 4.3.5.1.

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3.1.2 Mixture of synthetic filament and horsetail hair. The mixture in class 2 brushes shall be composed of 30 to 50 percent by weight horsetail hair meeting the requirements in 3.1.1, and the remainder synthetic filament. The filament shall be styrene-acrylonitrile copolymer or polypropylene at the option of the manufacturer. The styrene-acrylonitrile copolymer shall be in accordance with class S, type II filament of H-S-951 and in addition shall be resistant to gasoline when tested as specified in H-S-951 for resistance to solvents. Polypropylene filament shall be in conformance with class P in H-S-951. The average diameter of the filaments shall be 0.009 ± 0.001 inch when determined as specified in 4.3.3.

3.1.3 Wood. Wood in the blocks of shoe brushes shall be high- or medium-density hardwood such as beech, birch, maple, ash, or oak , as classified in Commercial Standard CS-35.

3.1.4 Staple wire. The staple wire shall be round carbon-steel wire, composition SAE 1006, 1008, 1010, or 1015, cold-drawn or annealed in process and finished with any one of the finishes in QQ-W-461.

3.2 Design. The brushes shall be of the general designs shown in figure 1. The figures are for illustration purposes and are not intended to preclude brushes otherwise meeting the requirements of this specification. The brushes shall be designed to be held firmly and easily in the hand.

3.3 Construction.

3.3.1 Type I, dauber brush. The dauber block shall be a single piece of flat wood smoothly finished, rounded at the ends, and free from sharp edges. The brush part shall be circular curving into a shaped handle. The holes for the tufts shall be uniform and shall be evenly distributed and symmetrically located in the circular face of the block. The holes shall be completely filled with tufts of hair (class 1) or hair and filament (class 2) evenly trimmed.

3.3.2 Type II, polishing brush. The block of the polishing brush shall be a single piece of flat wood rectangular-shaped and rounded at the ends, with finger grooves along each side. The block shall be smoothly finished and free from sharp edges. The holes for the tufts shall be uniform and shall be evenly distributed and symmetrically located in the face of the block. The holes shall be completely filled with tufts of hair (class 1) or hair and filament (class 2) evenly trimmed.

3.4 Coating. Blocks shall be evenly coated with a suitable transparent finish such as wax, varnish, or lacquer.

3.5 Physical requirements. The physical requirements of the brushes shall be as specified in table I.

TABLE I. Minimum physical requirements of brushes

Brush component	Type I, dauber brush	Type II, polishing brush
Block face:		
Length ---------- inches --	5-7/8	------
At center ---------- do. --	-----	7-1/8
At sides ---------- do. --	-----	6-7/16
Width ---------- inches --	-----	2-3/16
Circular face ---------- do. --	1-3/4	------
Handle portion:		
At widest point ------ inch ---	1	------
At narrowest point --- do. ---	11/16	------
Thickness of block ---------- inch ---	1/2	1/2
Holes for tufts:		
Number ----------	31	116
Depth ---------- inch ---	5/16	5/16
Distance from edge ------- do. ---	1/8	3/16
Hair, or hair and filament:		
Length clear of block ---- inch ---	see note "a"	see note "a"
Weight per brush ---------- oz. ---	0.277	1.50
Staple wire, diameter ------ inch ---	0.0286	0.0348

a. The length of hair or filament clear of block shall be 1 ± 1/8 inches.

3.6 Retention of tufts. When subjected to a pulling force of 8 ± 0.5 pounds, as described in 4.3.1 none of the tufts tested shall be dislodged from the block.

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3.7 Weight of hair (class 1) or hair and filament (class 2). The minimum weight of the hair, or hair and filament, as applicable, shall be as specified in table I when determined as described in 4.3.2.

3.8 Ductility of wire. When the loop formed by the staple is tightened until the resultant opening is no larger than the diameter of the staple wire, as described in 4.3.4, the steel wire shall not fracture.

3.9 Identification marking. Brushes shall be marked with the manufacturer's name or trademark. In addition brushes shall be marked or tagged with the name of the hair and filament used in the brush part.

3.10 Workmanship. Brushes shall be free from defects which may affect their durability, serviceability or appearance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling for inspection and acceptance. Sampling for inspection and acceptance shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Component and material inspection. In accordance with 4.1 above, the supplier is responsible for insuring that components and materials used are manufactured, tested, and inspected in accordance with requirements of referenced subsidiary specifications and standards to the extent specified, or if none, in accordance with this specification.

4.2.1.1 Process examination of sterilization of horsetail hair. In process examination shall be made to determine that horsetail hair is sterilized by one of the following methods.

Method A, boiling water. The horsetail hair shall be completely submerged in a container of water maintained at a temperature not less than 100°C. (212°F.) for 30 minutes.

Method B, dry heat. The hair shall be placed in an oven and the temperature of the oven brought to and maintained at a temperature not less than 140°C. (284°F.) and not more than 145°C. (293°F.) for 3 hours.

Method C, autoclaving. The hair shall be exposed to steam under pressure in an autoclave at a minimum temperature of 120°C. (248°F.) for 20 minutes. The steam shall be measured in the exhaust line at its exit from the autoclave by an indicating thermometer which is accurate within ± 1°C. (1.8°F.) and by a recording thermometer adjusted to read no higher at any time than the indicating thermometer. The time to exposure shall be measured from the moment at which the indicating thermometer reaches the specified sterilization temperature. The hair shall be sterilized in tied or wrapped bundles not exceeding 2-1/2 inches in diameter with the end of the bundles open, or in untied or unwrapped lots not exceeding 2-1/2 inches in depth. The bundle shall be placed on racks or trays in single layers with the trays or racks separated from each other sufficiently to assure free circulation of the steam and exposure of all the hair to such steam.

4.2.1.2 Intermediate testing. Test shall be made on each lot of wood offered for fabrication into blocks to determine conformance with 3.1.3. Method B of ASTM D 2016 shall be used. The inspection level shall be S-2, acceptable quality level (AQL) 2.5 defects per 100 units. Three determinations shall be made on each piece of wood. The sample unit shall be one piece of wood.

4.2.2 End item examination and testing.

4.2.2.1 Lot. A lot shall consist of all brushes of one type and class offered for examination and testing at the same time.

4.2.2.2 Visual examination. Brushes shall be examined visually for the defects in table II. The inspection level shall be II, acceptable quality level (AQL) 4.0 defects per 100 units. The unit for examination shall be one brush. If the number of defects in the sample for visual examination exceeds the acceptance number in MIL-STD-105, the entire lot of brushes shall be considered as having failed to meet the requirements of the specification.

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TABLE II. Visual defects

Brush component	Defects
Block	Not seasoned, dense, close-grain hardwood smoothly finished and in one solid piece.
	Splinters, slivers, cracks, splits, checks, hole knots, loose knots, tight or firmly encased knots 3/16 inch or more in diameter, worm holes, and doze.
	Finger grooves absent.
	Less than specified number of holes.
	Not coated as specified (see 3.4).
	Coating tacky.
Hair or filament	Not evenly distributed.
	Tufts not symmetrically arranged.
	Loose hair or filament.
	Loose tufts.
	Tuft missing.

4.2.2.3 Nondestructive dimensional examination. Brushes shall be examined for conformance with the applicable dimensional requirements in table I, with the exception of depth of holes and diameter of the staple wire which require destruction of the brush (see 4.2.2.4). The inspection level shall be S-3, acceptable quality level (AQL) 6.5 percent defective. The unit for examination shall be one brush. If any brush in the sample fails to meet any of the specified dimensional requirements in table I, the brush shall be considered defective. If the number of defective brushes in the sample exceeds the acceptance number in MIL-STD-105, the entire lot of brushes shall be considered as having failed to meet the requirements of this specification.

4.2.2.4 Destructive dimensional examination. Brushes shall be examined for conformance with the requirements in table I for depth of holes and diameter of staple wire. The inspection level shall be S-2, acceptable quality level (AQL), 4.0 percent defective. The unit for examination shall be one brush. If any brush in the sample fails to meet any of the requirements for which examination is made, the brush shall be considered defective. If the number of defective brushes in the sample exceeds the acceptance number in MIL-STD-105, the entire lot of brushes shall be considered as having failed to meet the requirements of this specification.

4.2.2.5 Testing. Brushes shall be tested for conformance with the applicable requirements of this specification by the methods specified in 4.3. The test unit shall be two brushes. The inspection level shall be S-2, acceptable quality level (AQL) 4.0 percent defective. If any brush in the sample for test fails to meet any of the requirements for which tests are made, the test unit shall be considered defective. If the number of defective test units exceeds the acceptance number in MIL-STD-105, the entire lot of brushes shall be considered defective.

4.3 Test methods. Unless otherwise specified herein, tests shall be conducted at room temperature, $27^{\circ} \pm 5^{\circ}C.$ ($80^{\circ} \pm 9^{\circ}F.$).

4.3.1 Tuft retention. Five tufts from different locations on one brush shall be tested as follows: Secure the block in a suitable stationary clamping device and the tuft ends in another clamping device so that the hairs and filaments are evenly distributed and so as to prevent distortion and cutting. Apply a pulling force of 8 ± 0.5 pounds to the tuft in a direction perpendicular to the face of the block to determine conformance with 3.6.

4.3.2 Weight of hair and filament. The hair from one class 1 brush or the hair and filament from one class 2 brush, as applicable, shall be removed from the block, conditioned for not less than 16 hours in air at a temperature of $23^{\circ} \pm 1.1^{\circ}C.$ ($73.5^{\circ} \pm 2^{\circ}F.$) and a relative humidity of 50 ± 4 percent, and weighed in the same atmosphere. The results shall be checked for conformance with the requirements in table I.

4.3.3 Diameter of synthetic filaments. Thirty filaments shall be taken at random from the specimen used to make the test in 4.3.2. The average diameter of the filaments shall be determined as specified in H-S-951 for conformance with 3.1.2.

4.3.4 Ductility test for wire. Three tufts shall be removed from the brush and the staples separated from the hair or filament. The loop formed by the staples shall be tightened by means of pliers or other suitable means until the resultant opening is no larger than the diameter of the wire, and the wire examined for conformance with 3.8.

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4.3.5 Dispute testing.

4.3.5.1 Identification of horsetail hair. The hair for identification shall be taken at random from the specimen used to make the test in 4.3.2. Not less than 115 hairs are required to make the tests, (a) at least 15 for cross sectional examination and (b) at least 100 for diameter measurement by microprojection, not less than 15 of which shall be examined at the same time for conformance with the significant longitudinal characteristics of horsehair. Prior to testing, the hairs shall be boiled for at least 15 minutes to remove any foreign materials that could obscure significant characteristics and to facilitate sectioning.

(a) Cross sectional examination. Cross sections of not less than 15 hairs shall be prepared by a suitable sectioning device, mounted on slides as specified in ASTM D 276, and examined for conformance with the significant cross sectional characteristics of horsehair in table I of ASTM D 276.

(b) Longitudinal examination and diameter measurements. At least 100 hairs shall be prepared by means of a suitable sectioning device, and segments, one from each hair, mounted on slides as specified in ASTM D 2130. Not more than 50 segments shall be mounted on one slide. At least 15 segments shall be examined for conformance with the significant longitudinal characteristics of horsehair in table I of ASTM D 276. All 100 segments shall be microprojected as specified in ASTM D 2130 and the diameter of each segment measured, recorded, and the average diameter calculated. The minimum, maximum, and average values shall be checked for conformance with 3.1.1.

4.3.5.2 Weight of horsetail hair in class 2 brushes. The percent horsetail hair by weight in class 2 brushes shall be determined as described in method 2101 of Fed. Test Method Std. No. 191 except that the specimen shall be the hair and filament from one brush, and foreign material shall be removed by agitating the specimen in a 1 percent neutral soap solution at a temperature of $38^{\circ} \pm 3^{\circ}$C. ($100^{\circ} \pm 5.6^{\circ}$F.), rinsing by agitating the specimen in water at the same temperature until the water is clear, and draining to remove surplus water. The percent horsetail hair by weight in the specimen shall be calculated on the basis of the weight of the washed and dried specimen (S).

4.3.5.3 Identity of filament in class 2 brushes. Sufficient tufts shall be removed from one class 2 brush and separated into filament and hair to provide enough filament for the required tests. Acrylonitrile-styrene (class S) or polypropylene (class P), as applicable, shall be tested for conformance with the following requirements applicable to each class and as specified in H-S-951: Specific gravity, melting point or softening point, resistance to solvents, hydrochloric acid, and n-butanol. In addition, acrylonitrile-styrene filaments shall be tested for resistance to gasoline and examined for conformance with 3.1.2.

5. PREPARATION FOR DELIVERY

5.1 Packaging. The brushes shall be level A, B, or C as specified (see 6.2).

5.1.1 Level A or B. Each set consisting of two (2) brushes, or twelve (12) individual brushes of one description shall be packaged in a close-fitting box conforming to PPP-B-566, PPP-B-665, or PPP-B-676. Each box shall contain 3 balls or the equivalent in flakes of naphthalene conforming to R-N-91, to protect the brushes from moths.

5.1.2 Level C. The brushes shall be packaged in accordance with the supplier's commercial practice.

5.1.3 Standard pack (civil agency procurements). The standard pack for civil agency procurements shall be as specified in paragraph 5.1.1.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Twenty-four (24) sets (48 brushes) consisting of two (2) brushes each or twelve (12) packages (144 brushes) of brushes shall be packed in a close-fitting box conforming to PPP-B-636, class weather resistant. The box shall be waterproofed with tape and strapped in accordance with the appendix to the box specification.

5.2.2 Level B. The brushes shall be packed as specified in 5.2.1 except that the box shall be class domestic and waterproofing with tape will not be required.

5.2.3 Level C. The brushes, packaged as specified in 5.1, shall be packed to insure carrier acceptance and safe delivery at destination at lowest rates, in containers complying with the Uniform Freight Classification Rules or National Freight Classification Rules, as applicable.

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5.3 Marking.

5.3.1 In addition to any special marking required by the contract or order, interior and shipping containers shall be marked in accordance with Fed. Std. No. 123 or MIL-STD-129, as applicable.

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents.

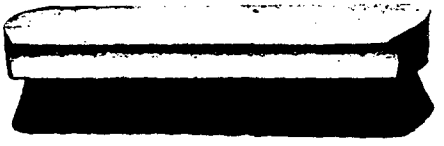
(a) Title, number, and date of this specification.
(b) Type and class required (see 1.2).

6.2 Standard pack for civil agencies. The standard pack requirements in 5.1.1 are intended for use in procurements of stores stock replenishments. Procuring officers should use the standard pack requirements when it is known that the material will be shipped from a supplier to a domestic warehouse, supply depot or intermediate storage point for temporary storage, subsequent issue or shipment to eventual user.

DOD coordination has been waived.



Type I. Dauber brush



Type II. Polishing brush

Figure 1. Shoe brushes

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GENERAL SERVICES ADMINISTRATION - FEDERAL SUPPLY SERVICE

SPECIFICATION COMMENT SHEET

BUDGET BUREAU NO. 29-R0175

INSTRUCTIONS

This form provides a way for users of this specification to inform the originator of problems encountered in its use. It is not to be used to request changes to accommodate proprietary features. All comments will be considered and appreciated, but please do not expect a reply. To comment: detach, complete, fold, staple, and mail.

NOTE: Comments on this form do not constitute or imply authorization to waive any part of the document or serve to amend contractual requirements.

1. SPECIFICATION

H-B-00581A(GSA-FSS) Brushes, Shoe

2. CONTRACT NO. *(If any)*	3. QUANTITY ON CONTRACT *(Optional)*	4. DOLLAR VALUE *(Optional)*

5. GENERAL NATURE OF PROBLEM *(e.g., inspection difficulties, manufacturers unable to meet tolerances, containers collapse under normal warehousing conditions, etc.)*

6. SPECIFIC REQUIREMENTS AFFECTED *(Include paragraph number and lines of wording)*

7. SPECIFIC PROBLEMS *(e.g. tests in 4.2.2 will not assure that the battery will last required time; temperature ranges in table 2 do not conform to commercially available items.)*

8. RECOMMENDATIONS

9. NAME OF MANUFACTURER, ASSOCIATION, GOVT., AGENCY, ETC.	10. ADDRESS *(Number, Street, City, State and Zip Code)*

11. NAME AND TITLE OF SUBMITTER	12. DATE

FORM GSA FEB 69 2200

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